

Assignment 2 Q3: SVM and Logistic Regression used for Regression

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1. Support Vector Regression (SVR) Example

Given data:

x	y
1	2.1
2	2.9
3	3.8
4	4.9
5	5.1

Step 1: Fit linear SVR:

$$f(x) = wx + b$$

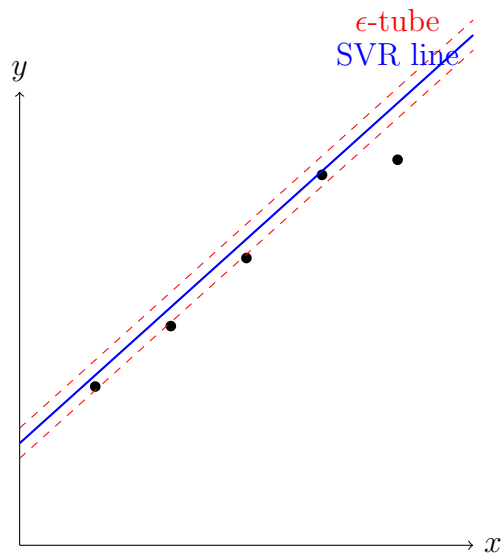
$$w \approx \frac{5.1 - 2.1}{5 - 1} = 0.75, \quad b \approx 2.1 - 0.75 \cdot 1 = 1.35$$

$$f(x) = 0.75x + 1.35$$

Step 2: Predict for $x = 2.5$:

$$\hat{y} = 0.75 \cdot 2.5 + 1.35 = 3.225 \approx 3.23$$

SVR -tube Illustration:



2. Logistic Regression Regression-like Example

Given data:

x	y (probability)
1	0.2
2	0.4
3	0.6
4	0.8

Logistic regression model:

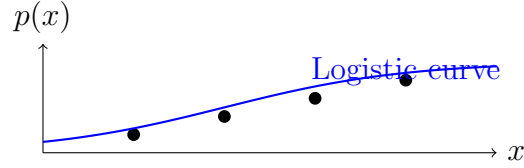
$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}, \quad \beta_0 = -2, \beta_1 = 1$$

$$p(x) = \frac{1}{1 + e^{-(x-2)}}$$

Predict for $x = 2.5$:

$$p(2.5) = \frac{1}{1 + e^{-0.5}} \approx 0.622$$

Logistic Curve Illustration:



3. Summary Table

Method	Input x	Output y	Prediction at $x = 2.5$
SVR (linear)	1,2,3,4,5	2.1,2.9,3.8,4.9,5.1	3.23
Logistic Regression	1,2,3,4	0.2,0.4,0.6,0.8	0.622

Summary of Research Article: Hybrid SVM-Logistic Regression for COVID-19 Outcome Prediction

In this study, Support Vector Machine (SVM) and Logistic Regression (LR) were applied as individual classifiers and later combined into a **hybrid model** to improve prediction accuracy of COVID-19 outcomes, particularly when the number of events per variable (EPV) is small.

1. Support Vector Machine (SVM)

- **Purpose:** Binary classification (e.g., predicting death vs. recovery from COVID-19).
- **Kernel:** Linear kernel used for simplicity and efficiency.
- **Mechanism:** SVM finds the optimal hyperplane that maximizes the margin between classes:

$$f(x) = w^T x + b$$

Non-linear cases use a kernel function, but here a linear kernel was applied.

- **Advantage:** Effective in high-dimensional spaces, less prone to overfitting for low EPV.
- **Role in Study:** Trained on COVID-19 data (confirmed cases, recoveries, deaths) and evaluated using accuracy, MSE, and RMSE.

2. Logistic Regression (LR)

- **Purpose:** Predicts the probability of an event (e.g., death from COVID-19).
- **Model Form:**

$$y(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_j x_p + \varepsilon_i)}}$$

where $y(x)$ is the probability of the event.

- **Threshold:** If $y(x) \geq 0.5$, outcome = 1; else 0.
- **Limitation:** LR can produce unstable or biased coefficient estimates with low EPV.
- **Role in Study:** Applied to same dataset; performance compared with SVM and the hybrid model.

3. Hybrid SVM-LR Model

- **Objective:** Combine strengths of SVM and LR, overcome individual limitations.
- **Approach:**
 1. Train both SVM and LR on the same dataset.
 2. For each data point, select the better prediction from either SVM or LR.
 3. Combine outputs to form the final prediction.
- **Advantages:**
 - Improved accuracy, lower MSE and RMSE.
 - More stable performance across different EPV values (2,3,4,5).

4. Evaluation Metrics

- **Accuracy:** Proportion of correct predictions.
- **MSE (Mean Squared Error):** Average squared difference between actual and predicted values.
- **RMSE (Root Mean Squared Error):** Square root of MSE for interpretability.

5. Key Findings and Conclusion

- Hybrid model outperformed both SVM and LR individually:
 - Higher accuracy.
 - Lower MSE and RMSE.
 - More stable coefficient estimates with small EPV.
- Particularly useful in medical decision-making during pandemics.
- Demonstrates that hybrid SVM-LR can effectively improve outcome prediction with limited data, making it valuable for public health planning.

Summary of Research Article: Multiple Linear Regression for COVID-19 Forecasting

In the paper by Semwal, Bahuguna, Uniyal, and Vyas (2023), the authors applied a multiple linear regression (MLR) model to forecast daily active COVID-19 cases in Uttarakhand, India.

Objectives and Data

- Use secondary data from the COVID-19 health bulletin of Uttarakhand.
- Identify significant predictors of daily active cases and visualize trends of active, positive, and recovered cases.

Modeling Approach

- The authors build a Multiple Linear Regression model, modelling daily active cases as a linear function of several COVID-related variables.
- They compare a full MLR model (with more predictors) to a simpler linear regression version.

Results

- The MLR model achieves $R^2 = 0.90$, while the simpler linear regression model gets $R^2 = 0.88$.
- Significant predictors (p less than 0.001) for daily active cases:
 - Number of daily positive cases
 - Number of cured/recovered cases
 - Number of deceased cases
- Forecast: assuming the trend continues, the cumulative active cases on the 93rd day (since September 18, 2022) is projected to be approximately 6,695 (95 percent CI: 6,259 – 7,131).

Conclusion

- Regression models (especially MLR) are effective for forecasting COVID-19 active cases.
- Such forecasts can help health authorities and governments with planning, resource allocation, and policy-making in response to the pandemic.

Reference: Semwal J., Bahuguna A., Uniyal A., Vyas S. “A Study to Analyse Covid-19 Outbreak Using Multiple Linear Regression: A Supervised Machine Learning Approach.”